	MNLI	QNLI	QQP	RTE	SST	MRPC	CoLA	STS
RoBERTa$_{\text{BASE}}$								
+ all data + 500k steps	87.6	92.8	91.9	78.7	94.8	90.2	63.6	91.2
RoBERTa$_{\text{LARGE}}$								
with BOOKS + WIKI	89.0	93.9	91.9	84.5	95.3	90.2	66.3	91.6
+ additional data (§3.2)	89.3	94.0	92.0	82.7	95.6	**91.4**	66.1	92.2
+ pretrain longer 300k	90.0	94.5	**92.2**	83.3	96.1	91.1	67.4	92.3
+ pretrain longer 500k	**90.2**	**94.7**	**92.2**	**86.6**	**96.4**	90.9	**68.0**	**92.4**

Table 8: Development set results on GLUE tasks for various configurations of RoBERTa.

Hyperparam	**RoBERTa$_{\text{LARGE}}$**	**RoBERTa$_{\text{BASE}}$**
Number of Layers	24	12
Hidden size	1024	768
FFN inner hidden size	4096	3072
Attention heads	16	12
Attention head size	64	64
Dropout	0.1	0.1
Attention Dropout	0.1	0.1
Warmup Steps	30k	24k
Peak Learning Rate	4e-4	6e-4
Batch Size	8k	8k
Weight Decay	0.01	0.01
Max Steps	500k	500k
Learning Rate Decay	Linear	Linear
Adam ϵ	1e-6	1e-6
Adam β_1	0.9	0.9
Adam β_2	0.98	0.98
Gradient Clipping	0.0	0.0

Table 9: Hyperparameters for pretraining RoBERTa$_{\text{LARGE}}$ and RoBERTa$_{\text{BASE}}$.

Hyperparam	**RACE**	**SQuAD**	**GLUE**
Learning Rate	1e-5	1.5e-5	{1e-5, 2e-5, 3e-5}
Batch Size	16	48	{16, 32}
Weight Decay	0.1	0.01	0.1
Max Epochs	4	2	10
Learning Rate Decay	Linear	Linear	Linear
Warmup ratio	0.06	0.06	0.06

Table 10: Hyperparameters for finetuning RoBERTa$_{\text{LARGE}}$ on RACE, SQuAD and GLUE.